Weather, Interconnections, and Energy Prices

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Abstract

We evaluate pooled classical neural, graph and temporal controls across five training seeds and 49 market identities. A parameter-matched MLP, self-loop and shuffled graphs separate pooling from cross-zone messaging. Feature ablations and 168-hour GRU histories test additional explanations. Historical publication vintages remain unverified; results are exploratory.

1 Approach

At 12:00 Europe/Brussels on calendar day $D-1$, predict every hourly day-ahead price of delivery day D, including 23/25-hour daylight-saving days. We implement an independent dataset builder, per-zone XGBoost [1], and pooled hourly GATv2 models [2]. Directed graph edges are physical pairs observed during training; historical node lifetimes mask inactive identities. A shuffled control rewires non-self edges, preserving static in/out degrees without duplicates. Both GATs add self-loops, testing real topology under the same neural learner.

2 Experiments

Exact data and partitions. The master contains 5,887,954 zone-hours for 62 price-area identities, January 2016–October 2026. ENTSO-E prices, load forecasts and physical flows use releases v1.2. Additional sources are PECD/ERA5 temperature, anomaly and wind/solar capacity factors (2010–September 2026), and original CWE/CORE JAO exports (May 2015–October 2026). Weather is reanalysis, not archived forecasts; its requested v1.1 release identity is unverified. Missing values and negative prices are retained, and DE_AT_LU remains separate from DE_LU. Forty-nine identities have valid targets in fit, validation and test; the other 13 remain explicitly unscored, while all 62 can provide dated graph context.

Fit on January 1, 2016–March 31, 2024 (pretrain + train); validate on April 1, 2024–March 31, 2025; test on April 1, 2025–September 30, 2026, using Brussels delivery dates. October stays unassigned. MAE and RMSE are measured after inverse transformation, in EUR/MWh. Weekly persistence copies the same identity's price 168 elapsed hours earlier. Targets are never filled or filtered for feature completeness; neural preprocessing is fitted on training-period features only.

Controlled models. The prespecified study has 100 experiment/seed/topology combinations: four tabular references; MLP and real GAT with recent-price/calendar inputs, plus load alone, weather alone, JAO alone, or all groups; self-loop GAT and three fixed degree-preserving shuffled graphs; and GRU–MLP/GAT temporal pairs. Seeds are 224–228. MLP has two ELU/dropout layers, with width selected by parameter count only (96 for richer inputs, versus GAT's 64 channels/four heads). Models have no zone embeddings. Neural heads share training-only preprocessing, asinh(price/50), L1 loss, Adam 0.001, maximum 30 epochs and validation patience five. Histories contain 168 same-identity completed price intervals ending at Brussels noon, with explicit missingness. GRU has 32 units; both temporal heads share its inputs. XGBoost retains the prior depth-6/600-tree settings.

Multi-seed comparison. The richer-input controls below use common finite weekly-history hours. Values are mean $\pm$ sample standard deviation across five seeds, not prediction ensembles. Other feature sets, per-zone/country results, training-defined price-tail/season slices and rewiring variability are in the generated study appendix.

Model (richer inputs)	Validation MAE	Test MAE
Pooled XGBoost	20.74 ± 0.01	21.06 ± 0.01
Pooled MLP	21.44 ± 0.23	21.19 ± 0.21
Self-loop GAT	21.27 ± 0.09	21.06 ± 0.08
Real-graph GAT	20.98 ± 0.11	20.97 ± 0.13
GRU + MLP	24.37 ± 1.65	24.50 ± 1.22
GRU + real GAT	23.62 ± 0.52	23.59 ± 0.51
Shuffled GAT (3 graphs)	21.21 ± 0.05	21.45 ± 0.07

Table 1: Matched MAE (EUR/MWh), five seeds. Lower is better.

Graph contribution. Real-GAT minus pooled-MLP test MAE is -0.21 EUR/MWh; negative favours GAT. The paired validation MAE improvement for MLP to GAT is 0.46 (95% calendar-week bootstrap interval [0.20, 0.86]). Weeks are sampled jointly across countries after averaging fitted-model losses across seeds. This interval conditions on fitted models; seed and rewiring variability are reported separately. Self-loop GAT controls the architecture, pooled XGBoost/MLP control shared learning, and temporal pairs compare graph use with the same history encoder.

Earlier evidence. The earlier single-seed real GAT scored 20.83 versus 21.54 for shuffled GAT on matched test hours. France diagnostics motivated recent-price inputs: adding prior-day auction prices reduced MAE from 32.09 to 22.20. These earlier cohorts and learners are separate from this replication.

Timing and limitations. Previous-day prices are normally auctioned on $D - 2$ [3]; delivery is not publication time. DST guards exclude lag-24 target-day prices. Load, weather and JAO revision/publication vintages remain unverified, including retrospective ERA5 [4]. No observation after the noon origin enters temporal histories. Training-only topology is static and need not reconstruct historical commissioning dates. Models use identical target keys/masks, retain missing features and never fill targets. Test has been inspected previously: these are conditional exploratory comparisons, not a fresh confirmatory holdout. Multiple feature comparisons do not establish causal effects.

3 Future work

Validate conclusions on a genuinely new chronological period and archived operational forecasts. Investigate detailed JAO constraint nodes and edge attributes after assessing the pooled, topology, feature and temporal controls, keeping physical flow, scheduled exchanges and MaxBex distinct.

References

[1] T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. *KDD*, 2016. arXiv:1603.02754.

[2] S. Brody, U. Alon and E. Yahav. How attentive are graph attention networks? *ICLR*, 2022. arXiv:2105.14491.

[3] Nord Pool. Day-ahead auction timeline. Accessed November 2026.

[4] ECMWF. ERA5 global atmospheric reanalysis. Accessed November 2026.